Graduate Work – 3: Carrier-Based PWM Techniques for Two-Level and Multilevel Inverter Topologies

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Introduction

Modulation techniques play an essential role in shaping the output waveform of power electronic inverters. By controlling the switching of semiconductor devices, modulation determines the voltage quality, harmonic content, and overall performance of the converter. In two-level inverters, Sinusoidal Pulse-Width Modulation (SPWM) is widely used because it is easy to implement and produces a waveform that closely follows the desired sinusoid.

As converter technology progresses, multilevel inverters have become popular for applications that require improved waveform quality and reduced switching stress. These inverters rely on carrier-based modulation methods, with techniques such as In-Phase Disposition (IPD) and Alternative Phase Opposition Disposition (APOD) commonly used to manage multiple voltage levels.

In this work, MATLAB/Simulink is used to implement SPWM for a two-level inverter and IPD/APOD modulation for a four-level inverter. The simulations help visualize how each method forms voltage steps, influences harmonic distribution, and affects current waveforms. This study provides a clear comparison of basic and multilevel modulation strategies and highlights their impact on inverter output behavior.

Simulation Studies

1. SPWM Implementation for Two-Level Inverter

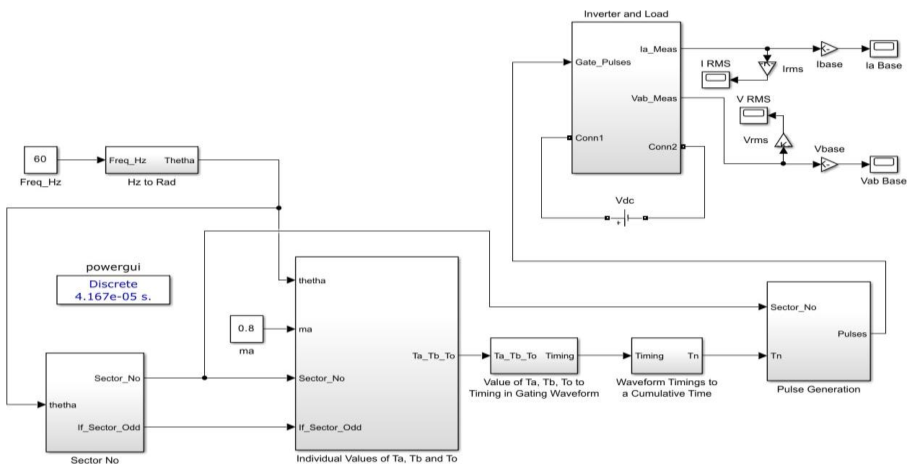

Fig.1 Simulink layout of the SPWM-controlled two-level inverter

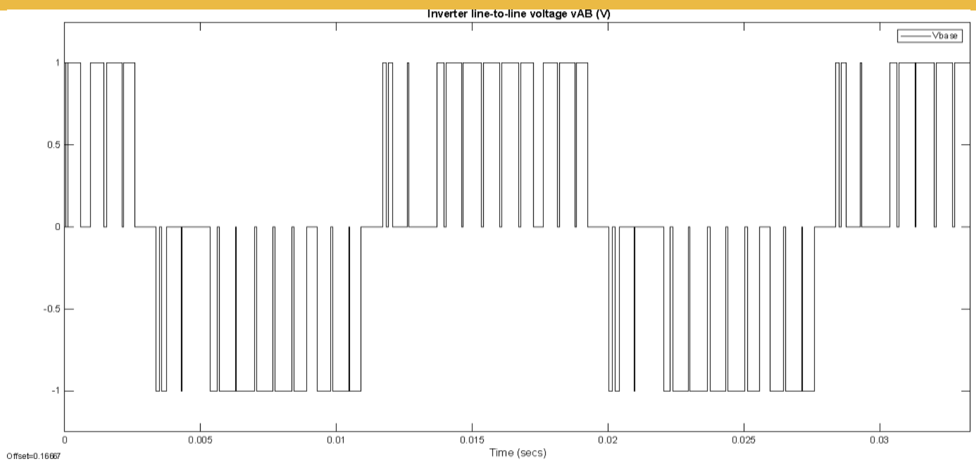

Fig.2 Generated line-to-line voltage waveform using SPWM

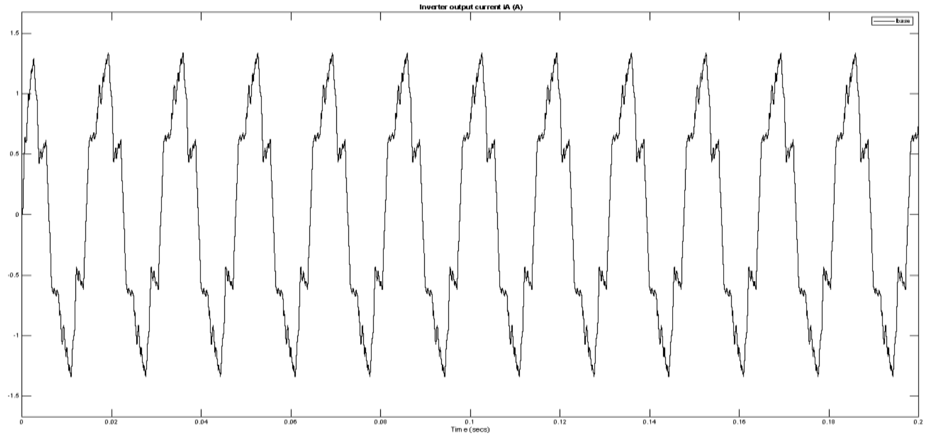

Fig. 3 Load current response under SPWM operation

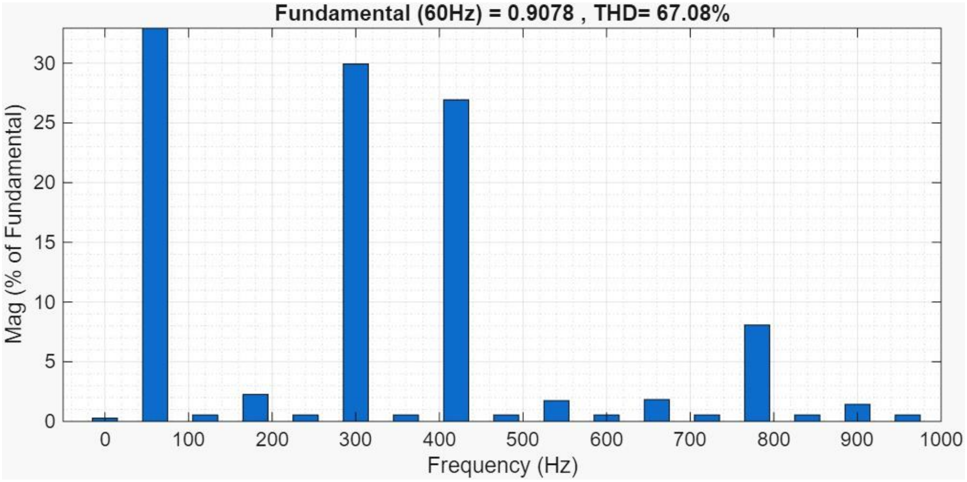

Fig.4 Harmonic spectrum of vAB using FFT analysis

2. Multilevel Modulation

A. Using IPD for Four-Level Inverter

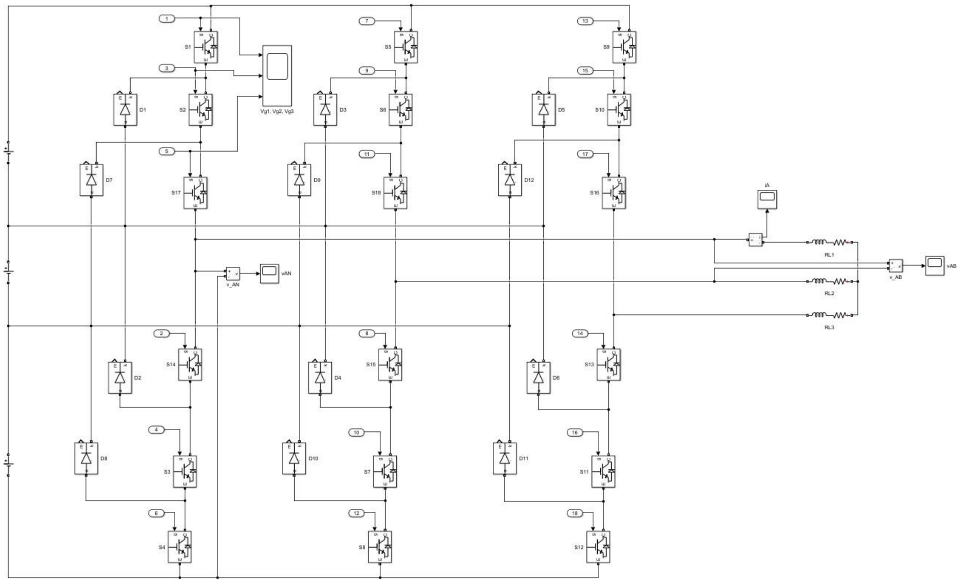

Fig.5 IPD-based four-level inverter model in Simulink

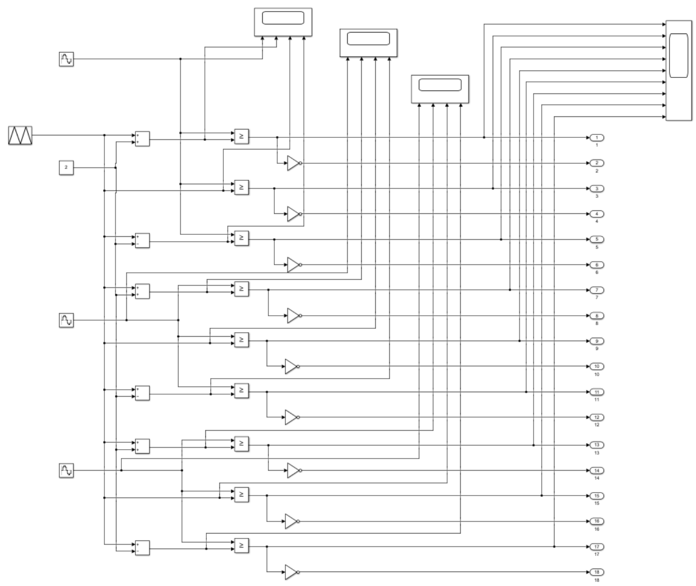

Fig.6 Carrier arrangement (Vcr1, Vcr2, Vcr3) used for IPD modulation

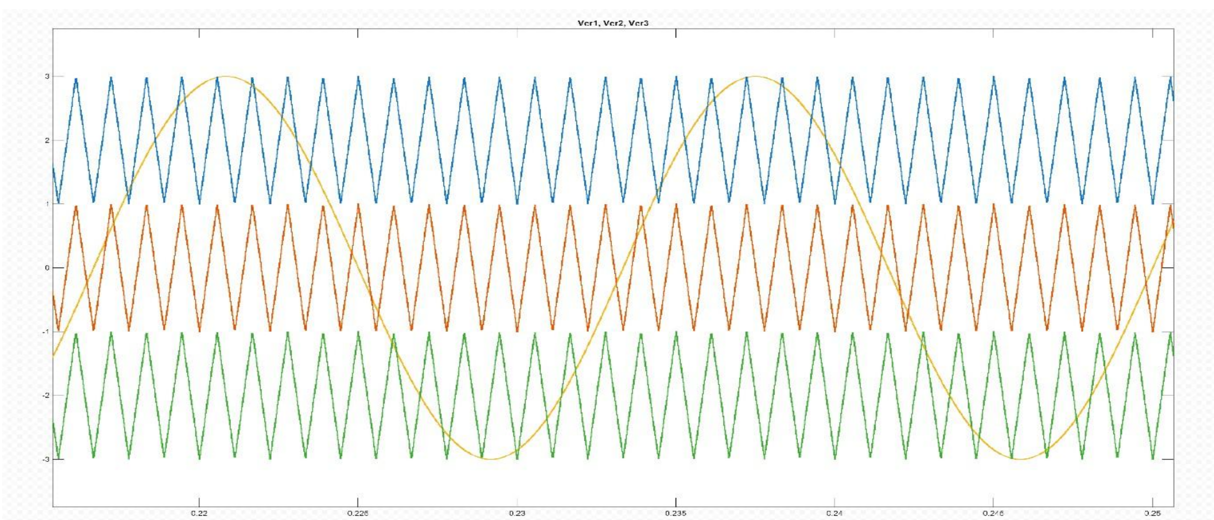

Fig.7 IPD carrier and reference signal interaction

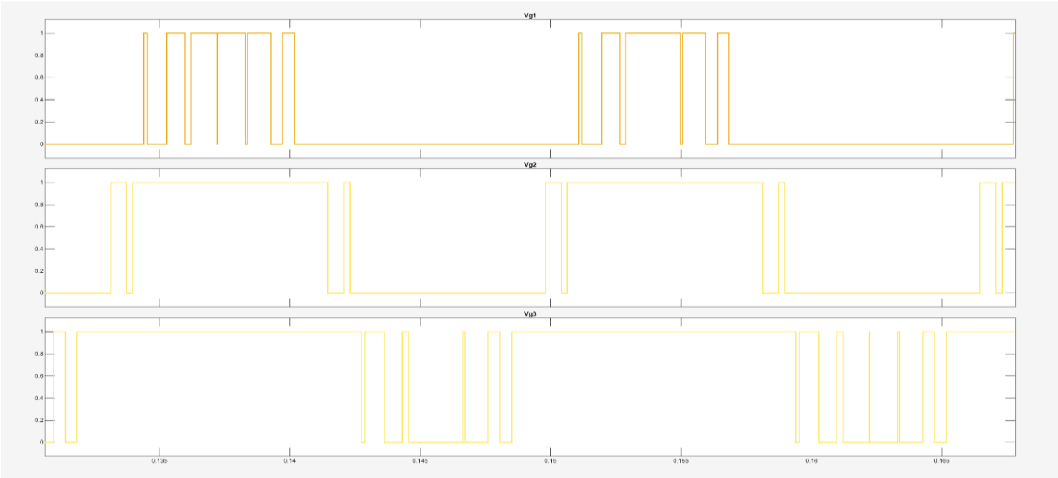

Fig.8 Gate pulse patterns (Vg1, Vg2, Vg3) generated through IPD

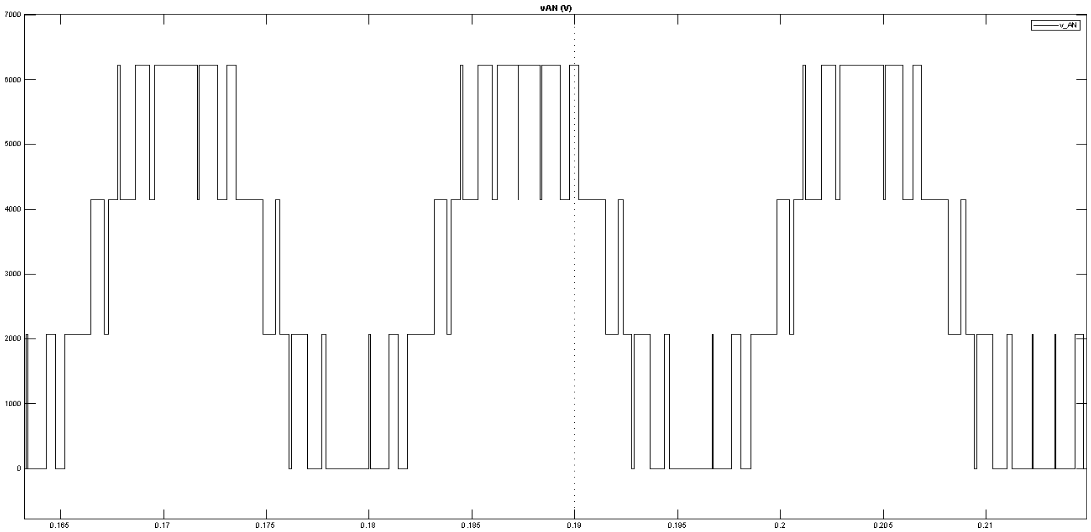

Fig.9 Output phase voltage vAN obtained from IPD strategy

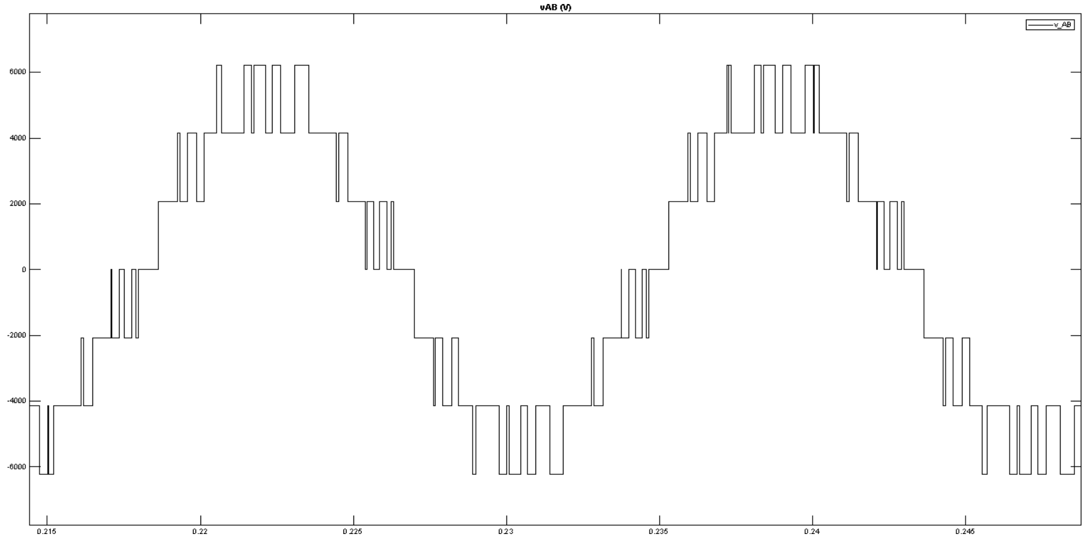

Fig.10 Line-to-line voltage vAB for IPD-modulated inverter

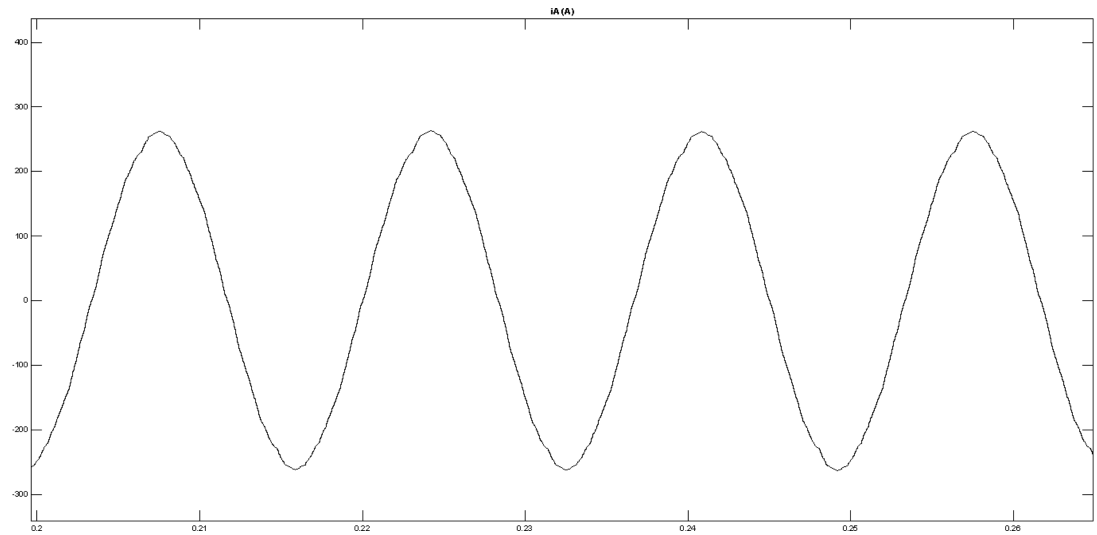

Fig.11 Load current waveform under IPD modulation

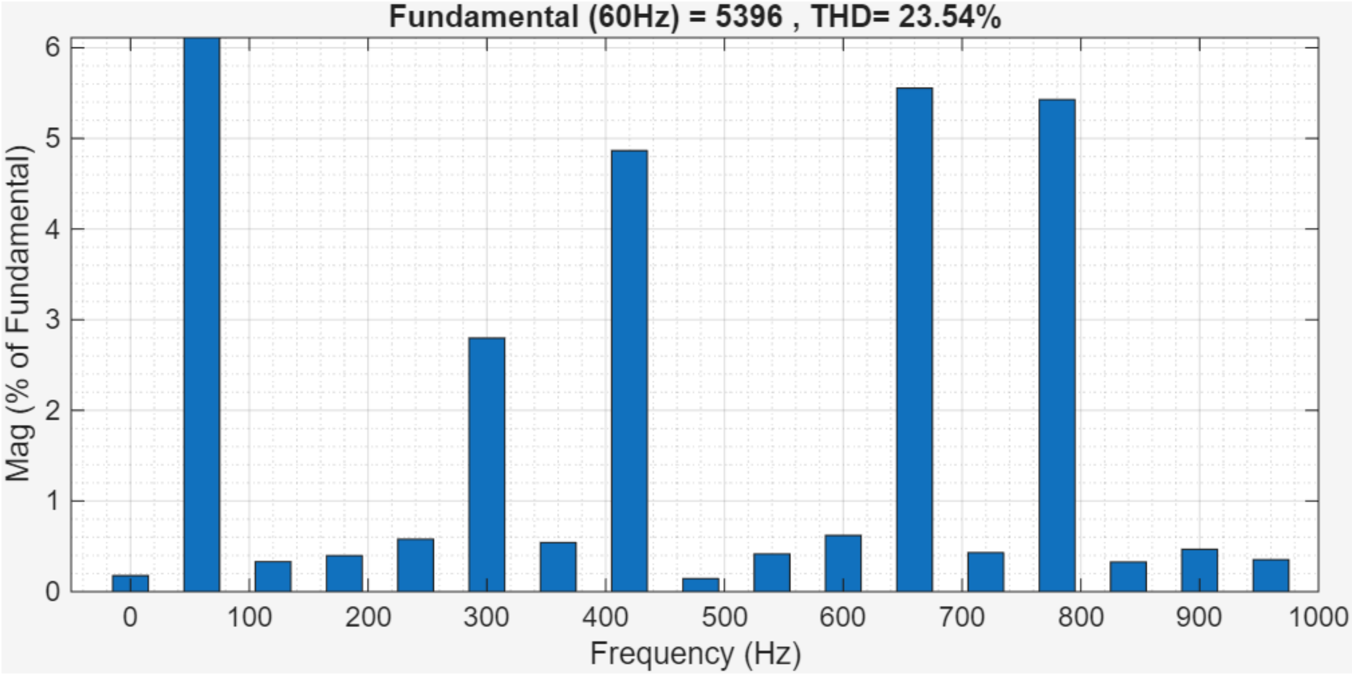

Fig.12 FFT spectrum of vAB showing harmonic performance under IPD

B. AOPD -Based Switching Strategy for Four-Level Inverter

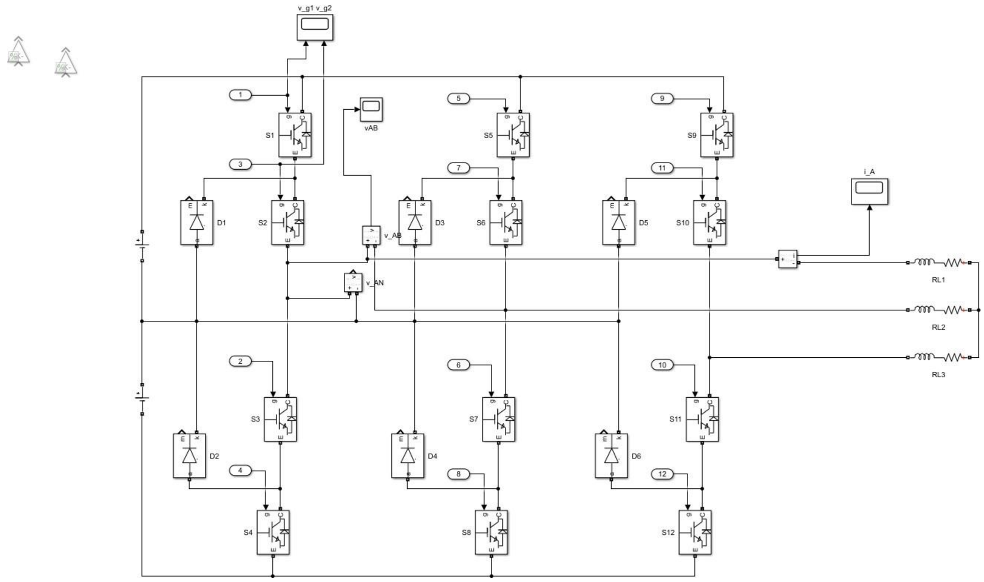

Fig.13 APOD-based four-level inverter model representation

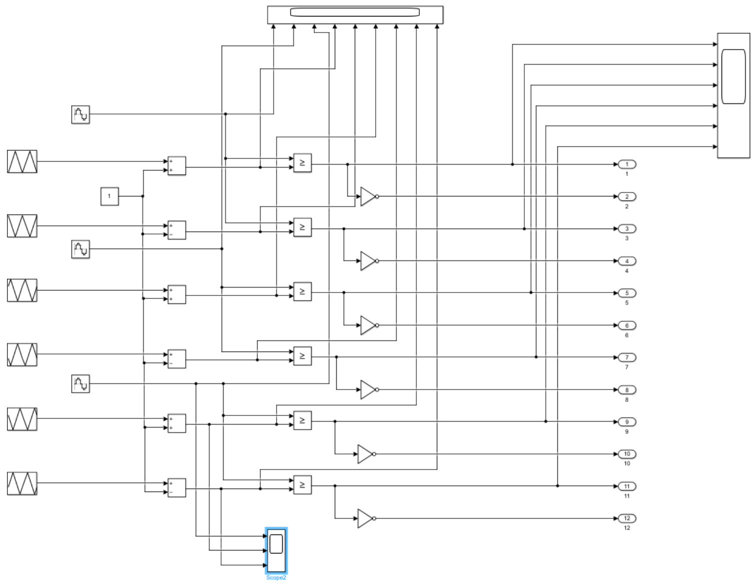

Fig.14 APOD carrier signal arrangement (Vcr1, Vcr2, Vcr3)

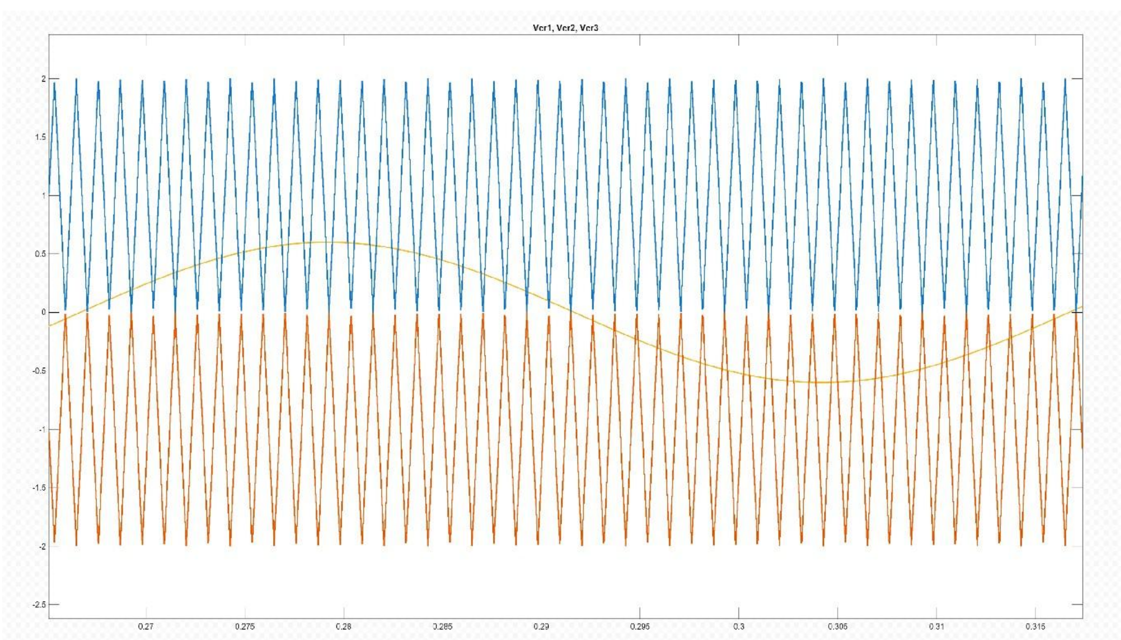

Fig.15 Comparison of APOD carriers with reference signal

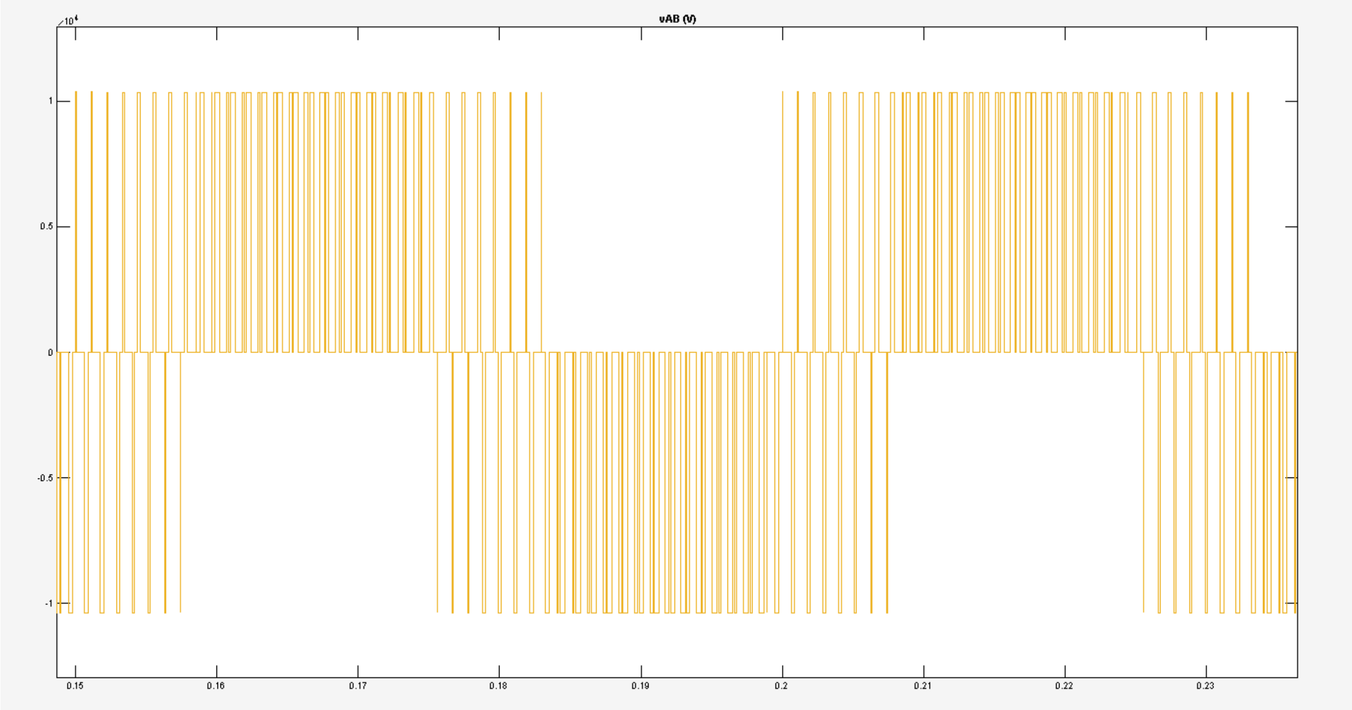

Fig.16 Output voltage vAB generated using APOD technique

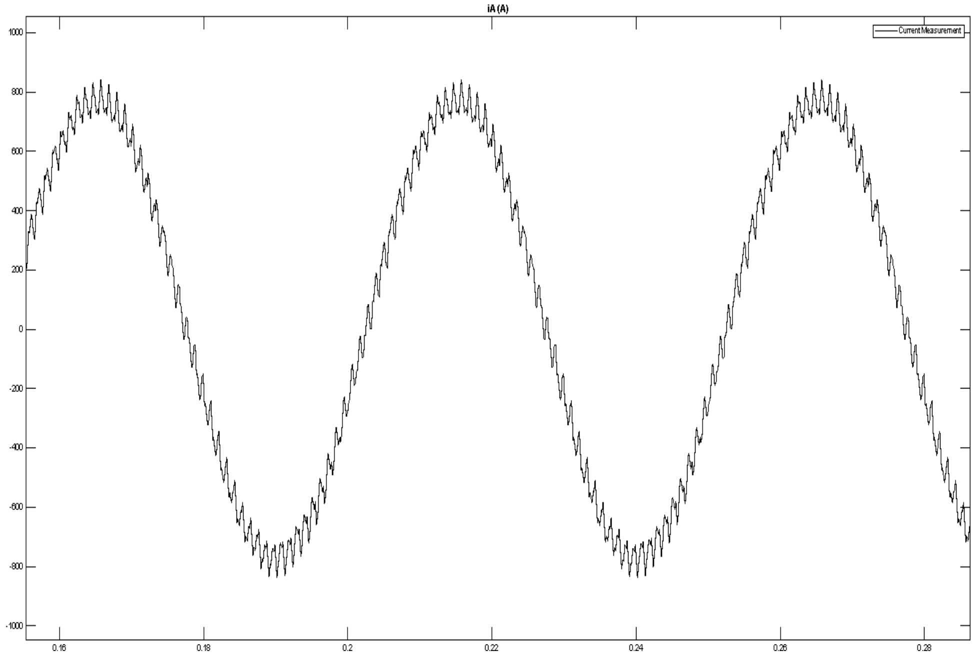

Fig.17 Load current (iA) for APOD-modulated operation

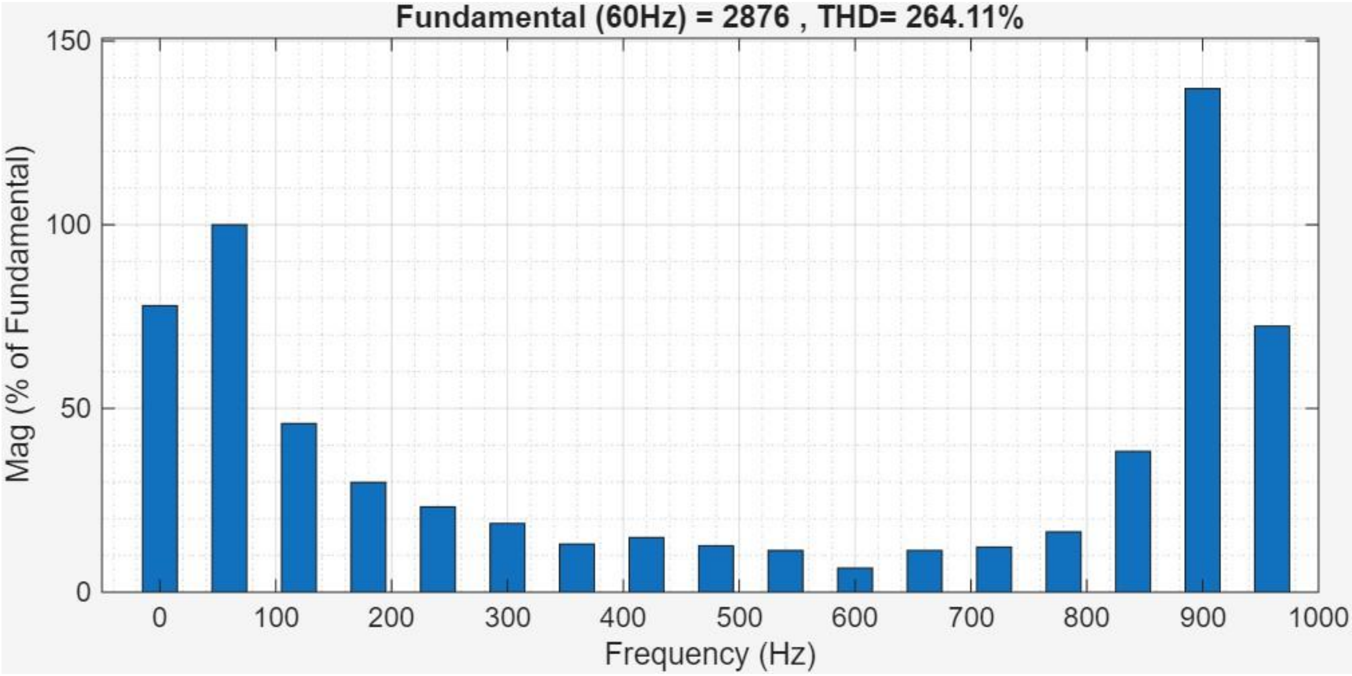

Fig.18 Harmonic spectrum of vAB for APOD method (FFT)

Conclusion

This study explored how different modulation strategies influence the performance of both two-level and multilevel inverters using MATLAB/Simulink models. The SPWM implementation for the two-level inverter demonstrated how a simple reference–carrier comparison can effectively synthesize a near-sinusoidal voltage and maintain predictable current behavior. This establishes SPWM as a reliable baseline technique for conventional converter systems.

The extension to multilevel inverters using IPD and APOD modulation highlighted the advantages of using multiple voltage levels for improved waveform quality. Both methods generated stepped output voltages characteristic of four-level inverters, with noticeable differences in switching patterns and harmonic distribution. IPD produced smoother transitions and more uniform switching behavior, while APOD introduced alternating carrier phases that redistributed harmonic content differently.

Overall, the simulations confirm that modulation choice significantly affects output quality, harmonic performance, and switching characteristics. By comparing SPWM with IPD and APOD, the study provides a clearer understanding of how modulation strategies scale from simple two-level inverters to advanced multilevel topologies. These insights support the design and optimization of high-performance converters used in modern applications such as electric drives, renewable energy systems, and power conditioning equipment.

References

MATLAB Documentation, "PWM Generation and Multilevel Inverter Modeling," MathWorks, Available: https://www.mathworks.com